Supplementary Material:
Gaussian Rank-Based Neighborhood Degree for Graph Neural Networks in Image Classification

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To distinguish supplementary content from the main paper, all sections, tables, and figures are prefixed with the letter "S" (e.g., Section S1, Table S1, Figure S1). This supplementary material is organized into the following sections:

1. **Efficiency Evaluation:** An evaluation of the execution time across the evaluated datasets.
2. **Graph Construction and Neighborhood Size:** A justification for setting the neighborhood size to $k = 40$, based on established configurations in previous literature.

S1 Efficiency Evaluation

We evaluated the computational efficiency by reporting the average training time, in seconds, for each dataset using the ViT-B16 feature extractor with the APPNP model, which corresponds to the most computationally demanding configuration. Results are reported both with and without GRaNDe in Figure S1.

Although the distances required by GRaNDe are recomputed at each training epoch, this operation is restricted to $n \cdot k$ elements, where n denotes the number of dataset images and k represents the neighborhood size. Since k is fixed, the computational cost scales linearly with dataset size, with an approximately $1.5\times$ overhead across datasets, ensuring that the proposed method remains efficient and scalable in practice.

All execution times were measured on a dedicated workstation with the following specifications:

- **OS & Kernel:** Pop!_OS 24.04 LTS (Linux kernel 6.17.9).
- **CPU:** Intel Core i9-13900K (13th generation).
- **RAM:** 32 GB DDR5.
- **GPU:** NVIDIA GeForce RTX 4070 Ti (12 GB VRAM).

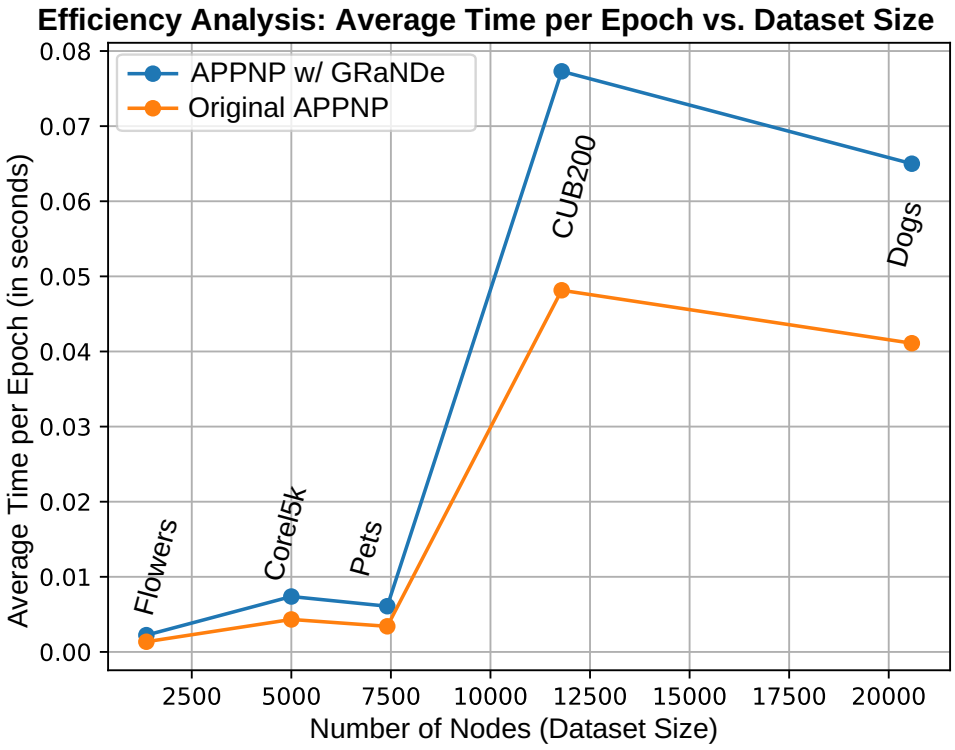


Figure S1: Impact of GRaNDe on training time (in seconds) across datasets.

S2 Graph Construction and Nerighborhood Size

The topological structure of the input graph plays an important role in the message-passing mechanism of GNNs. To build this structure, we first project the images into a latent space using the feature extractor and evaluate sample proximity via Euclidean distances, computed efficiently using a BallTree algorithm. Sorting these distances yields a ranked list of neighbors for each sample, reflecting local semantic similarities.

Formally, we construct an undirected graph $G = (V, E)$ over a node set $V = \mathcal{X} = \{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ representing the n dataset images. The edge set E is defined using a Reciprocal k-Nearest Neighbor (Rec. kNN) constraint, a well-established formulation in graph-based learning [1, 2]. Unlike standard kNN graphs, an edge is established if and only if two nodes mutually belong to each other's neighborhood, which naturally enforces sparsity and filters out asymmetric, noisy relations:

$$E = \{(i, j) \mid j \in \mathcal{N}(\mathbf{x}_i, k) \ \wedge \ i \in \mathcal{N}(\mathbf{x}_j, k)\}.$$

Regarding the neighborhood size, we fixed $k = 40$ across all experiments. This hyperparameter choice follows well-documented configurations in recent literature [1, 2, 3].

References

[1] J. Yang, H. Li, B. Du, and M. Ye, "Cheb-gr: Rethinking k-nearest neighbor search in re-ranking for person re-identification," in *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 19261–19270, 2025.

[2] L. P. Valem, D. C. Guimarães Pedronette, and L. J. Latecki, "Graph convolutional networks based on manifold learning for semi-supervised image classification," *Computer Vision and Image Understanding*, vol. 227, p. 103618, 2023.

[3] G. M. Brito and L. P. Valem, "Density-guided rank correlation graphs for graph convolutional networks in image classification," in *2025 38th SIBGRAPI Conference on Graphics, Patterns and Images (SIBGRAPI)*, pp. 1–6, 2025.